NOTE: The cover page of this Standard has been changed for administrative reasons. There are no other changes to this document. In addition, all dash sheets associated with this document are hereby designated as standard practices.

INCH-POUND

MIL-STD-1322A(NAVY)
26 MAY 1981
SUPERSEDING
MIL-STD-1322(NAVY)
14 August 1969

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

UNIT LOADS
OF
AMMUNITION AND EXPLOSIVES
FOR DOMESTIC AND
OVERSEAS SHIPMENT

AMSC N/A FSC 8140

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DEPARTMENT OF DEFENSE
Washington, DC 20362

Unit Loads of Ammunition and Explosives for Domestic and Overseas Shipment

MIL-STD-1322A

1 This Military Standard is approved for use by all Departments and Agencies of the Department of Defense The standard is published to establish requirements for the construction of unit loads of ammunition and explosives for domestic and overseas shipment which are not destined for underway replenishment (transfer-at-sea) or for unitized stowage aboard combatant ships.

2 As of the promulgation date of this document, this standard is a mandatory requirement to be invoked in all specifications, purchase descriptions, or military interdepartmental procurement requests for naval ammunition, explosives, and associated items intended for domestic and overseas shipment

3 MIL-STD-1322 (NAVY), dated 14 August 1969, was issued in book form and included a series of MIL STD dash numbered sheets MIL-STD-1322A supersedes the entire MIL-STD-1322, dated 14 August 1969, including the dash numbered sheets

Existing WR-53 slash sheets which have not been superseded shall be used until they are replaced by new MIL-STD dash numbered sheets

4 Requests for technical interpretations, approval of variations, deviations, of palletizing or unitizing plans or special assistance shall be sent to Commanding Officer Naval Weapons Station Earle, Colts Neck, NJ 07722, ATTN Naval Weapons Handling Center (NWHC) or call Autovon 449-7691, 7692 or 7693 or Area Code (201) 462-9500 Ext 691, 692 or 693

5 Copies of this standard and individual dash sheets may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120

6 Beneficial comments (recommendations, additions deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer Naval Ordnance Station Standardization and Documentation Division, Code 524, Indian Head Md 20640 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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FOREWORD

This Military Standard describes procedures and practices applicable to the construction of unit loads of ammunition, explosives and associated items for domestic and overseas shipping, handling and storage. This standard is not applicable for the construction of unit loads destined for underway replenishment (transfer-at-sea), amphibious operations or unitized stowage aboard combatant ships. These procedures are covered by MIL-STD-1323.

Physical dimensions, envelope shape, density, weight and type of items to be unitized and palletized vary greatly, making it difficult to address all possible configurations in one document. Therefore, the MIL-STD general document addresses the general requirements (design requirements, material, general palletizing principles, marking, inspection and test requirements). The MIL-STD dash number sheets provide specific approved instructions on constructing a unit load for a given item.

Existing WR-53 slash numbered sheets are being converted to MIL-STD-1322 dash numbered sheets on an as needed basis. When a WR-53 slash numbered sheet is revised, it will be assigned a MIL-STD-1322 dash number retaining the same identifying number as the WR slash numbered sheet. Until a superseding MIL-STD dash numbered sheet is published, the WR-53 slash numbered sheets form a part of this document and shall continue to be used.

The following cross reference is provided to facilitate identification between other WR documents that have been superseded by military standards.

NEW		PREVIOUS	
MIL-STD-1320	Truckloading of Ammunition and Explosives	WR-51	Shipping and Handling Instructions for Weapon System Components in Trucks and Trailers
MIL-STD-1322	Unit Loads of Ammunition and Explosives for Domestic and Overseas Shipment	WR-53	Palletizing Domestic Unit Loads
MIL-STD-1323	Unit Loads of Ammunition and Explosives for Underway Replenishment	WR-54	Palletizing Fleet Issue Unit Loads
MIL-STD-1325	Railcar Loading of Hazardous Materials	WR-52	Carloading of Weapons and Major Weapons System Components

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MILITARY STANDARD
UNIT LOADS OF AMMUNITION AND EXPLOSIVES FOR DOMESTIC AND OVERSEAS SHIPMENT

1 SCOPE

1.1 Scope This standard, consisting of a general document and dash numbered sheets, establishes the approved design and test methods for the preparation and construction of domestic and overseas unit loads of ammunition and explosives for shipping, handling and storage

1.1.1 MIL-STD-1322 MIL-STD-1322 is a dash numbered document that enables activities to prepare unit loads when a specific dash number sheet does not exist MIL-STD-1322 is limited to conventional ammunition or explosive items packaged in approved containers It is not authorized for loose or irregularly shaped items that require special purpose pallets or metal pallet adapters to construct the unit load

1.2 Application This standard is to be used by all personnel engaged in the construction of unit loads of ammunition and explosives intended for domestic and overseas shipment.

1 3 Restriction This standard shall not be used for unit loads of ammunition, explosives and associated items that are destined for underway replenishment (transfer-at-sea), amphibious operations or for unitized stowage aboard navy ships, except port to port cargo ships

2 REFERENCED DOCUMENTS

2 1 Issues of Documents The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein

SPECIFICATIONS

FEDERAL

FF-N-105	Nails, Brads, Staples and Spikes: Wire, Cut and Wrought
MM-L-751	Lumber, Softwood
NN-P-71	Pallets, Material Handling, Wood Stringer Construction, 2-Way and 4-Way (Partial)
NN-P-530	Plywood, Flat Panel
QQ-S-781	Strapping, Steel, and Seals

MILITARY

MIL-P-15011	Pallets, Material Handling, Wood Post Construction, 4-Way Entry

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MIL-P-23312	Pallet, Material Handling, Metal (For Ordnance Items) Mk 3 Mod 0, Mk 12 Mod 0 and Mk 12 Mod 1

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-731	Quality of Wood Members for Containers and Pallets
MIL-STD-1320	Truckloading of Ammunition and Explosives
MIL-STD-1323	Unit Loads of Ammunition and Explosives for Underway Replenishment
MIL-STD-1325	Railcar Loading of Hazardous Materials
MIL-STD-1660	Design Criteria for Ammunition Unit Loads

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

564200	Pallet, Material Handling Mk 3 Mod 0
2086479	Pallet, Material Handling Mk 12 Mod 0
2645217	Pallet, Material Handling Mk 12 Mod 1

PUBLICATIONS

HANDBOOK

MIL-HDBK-236	Index to Standards for Palletizing, Truck Loading, Railcar Loading and Container Loading of Hazardous Materials

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

OP 2165	Navy Transportation Safety Handbook for Hazardous Materials
OP 2173	Approved Handling Equipment for Weapons and Explosives
OP 4098	Handling Ammunition, Explosives and Hazardous Materials with Industrial Materials Handling Equipment (MHE)
OR-99/88 P1000	Pallet, Mk 3 Mod 0 and Pallet Mk 12 Mod 0

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(Copies of specifications, standards and publications may be obtained upon request from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

2.2 Other publications. The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the publication should cite "the latest issue and supplements thereto.")

3 DEFINITIONS

3.1 General. Definitions herein cover terms as they are used in this standard and are not to be confused with any definitions appearing elsewhere.

3.1.1 Ammunition. Components, and explosives in any case or contrivance prepared to form a charge, complete round, or cartridge for cannon, howitzer, mortar, or small arms; or for any other weapon, torpedo, mine, bomb, depth charge, demolition charge, fuze, detonator, projectile, grenade, guided missile, rocket and the like; signaling and illuminating pyrotechnic materials; explosive-loaded impulse devices such as bolts, squibs, and catapult charges, and all chemical warfare materials.

3.1.2 Approved palletizing and unitizing plan. A palletizing or unitizing plan developed in the absence of dash numbered sheets and approved for use by the Naval Weapons Handling Center (NWHC).

3.1.3 Batten. A wooden member used to fill space for the protection of lading against damage or to provide additional surfaces for strapping or bearing.

3.1.4 Cap. A cover with sides extending perpendicularly from its perimeter used as a protection of lading against damage, or to help create a stable load. It may be used over the load, inverted under a load or used under and over intermediate courses.

3.1.5 Dash number sheet. A MIL-STD dash numbered sheet (e.g., MIL-STD-1322-275) is one of a series of detailed instructions for the unitizing of specific ammunition and explosives or associated items.

3.1.6 DODIC or NALC. A four digit alpha/numeric code which will be either a Department of Defense Identification Code (DODIC) assigned by the Defense Logistics Services Center (DLSC) or a Navy Ammunition Logistic Code (NALC) assigned by the Navy Ships Parts Control Center, Mechanicsburg, PA.

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3.1.7 Dunnage. Lumber, strapping, nails or other material used to secure and protect unitized load

3.1.8 Edge protector. A thin piece of material used at the edge of a unit load to prevent damage by strapping.

3.1.9 Explosive The term explosives includes any chemical compound or mixture which, when subjected to heat, friction, detonation or other suitable initiation, undergoes a very rapid chemical change with the evolution of large volumes of highly heated gases which exert pressures in the surrounding medium

3.1.10 General document. The term general document is used to indicate those parts of this standard not identified by dash numbers.

3.1.11 Handling equipment Any device used to move packages, packs, unit loads, containers, items or components

3.1.12 Overhang. The distance that the vertical edges of the components of a unit load extend beyond the edges of the pallet

3.1.13 Pallet. A low portable platform of wood, metal or other suitable material used as a base for unit loads It is used as the base of a unit load to support and combine groups of commodities (or to confine single items) for handling and shipping as a single entity.

3.1.14 Pallet, special purpose A pallet which is specifically designed for use with a particular ammunition item or for use in a specific handling or transportation environment

3.1.15 Pallet adapter. A wood or metal framework designed to aid in securing irregularly shaped articles to a pallet

3.1.16 Partial unit loads A partial unit load is a unit load which holds or contains a quantity of items which is less than the number of items in a unit load constructed in accordance with the MIL-STD dash number sheet or approved palletizing/unitizing plan for the item.

3.1.17 Slash numbered sheet A Weapons Requirements (WR) slash numbered sheet (e.g., WR-53/275) is one of a series of detailed instructions for the unitizing of specific ammunition and explosives

3.1.18 Strapping. A length of flat steel, or other suitable material placed in tension around a unit load to secure the individual items into a single entity

3.1.19 Underhang The distance the edges of the pallet extend beyond the vertical edges of the items or containers stacked on it.

3.1.20 Unit load An assemblage of items (in or out of containers) designed to facilitate handling these items as a single entity

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4. GENERAL REQUIREMENTS

4.1 General. This section presents general requirements for unit loads of ammunition and explosives for domestic and overseas shipment which can be verified by visual observations, simple measurement, or by a certificate of compliance with material requirements. Inspection procedures are concerned only with those materials used to make up the unit loads, i.e., pallets, adapters, battens, strapping and the like. Inspection procedures for the ammunition or explosive items and their packaging are contained in documents relevant to the procurement or maintenance of the specific item. The MIL-STD dash numbered documents, the Weapons Requirements slash numbered documents that are being superseded and MIL-HDBK-236, an index to the slash numbered documents, form a part of this document.

4.2 MIL-STD dash numbered sheets. Specific instruction for constructing unit loads of ammunition and explosives for domestic and overseas shipment are provided by MIL-STD dash numbered sheets. Identification numbering of these MIL-STD dash numbered sheets consists of the basic MIL-STD-1322 designator followed by a dash number for each sheet. Assigned dash numbers, if cancelled, will not be used for other items and will become inactive.

4.2.1 Superseded WR-53 slash numbered sheets. When a WR-53 slash numbered sheet is revised, it will be assigned a MIL-STD-1322 dash number with the same identifying number as the WR slash sheet. It will supersede the WR-53 slash sheet previously in use. Until a superseding MIL-STD dash numbered sheet is published, the WR-53 slash numbered sheets form a part of this document. When unit loads for items not previously assigned WR-53 slash numbered sheets are developed, they will be assigned a MIL-STD-1322 dash number.

4.2.2 Use of MIL-STD dash numbered sheets. When a MIL-STD dash numbered sheet exists for a given item, the unit load shall be constructed in accordance with that sheet.

4.2.3 No MIL-STD dash sheet for item to be palletized. When required to palletize items for which a specific MIL-STD dash numbered sheet does not exist, MIL-STD-1322-000 shall be used. If there is a recurring requirement, the development of a specific MIL-STD dash numbered sheet should be requested. MIL-STD-1322-000 may be used only for those ammunition and explosive items that are packaged in approved containers and shall not be used for loose or irregularly shaped items requiring special purpose pallets or metal pallet adapters. Requests for guidance concerning items to be palletized should be addressed to Commanding Officer, Naval Weapons Station Earle, Colts Neck, NJ 07722, ATTN: Naval Weapons Handling Center (NWHC).

4.3 Index of dash numbered sheets. MIL-HDBK-236 titled "Index to Standards for Palletizing, Truckloading, Railcar Loading and Container Loading of Hazardous Materials" provides an index to MIL-STD-1322 dash numbered sheets, in addition to the other documents listed in the title. The handbook includes three types of listings:

Section 1 lists in alpha-numeric sequence, DODIC/NALC designated items that have dash/slash numbered sheets authorized for these items.

Section 2 lists in alphabetical sequence, all items that have dash numbered sheets authorized for those items.

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Section 3 lists all dash/slash numbered sheets numerically, giving the revision and change notice status of each sheet. In this listing, all MIL-STD dash numbered sheets are preceded by a dash (-) and all WR slash numbered sheets are preceded by a slash (/) enabling the user to determine whether the listed sheet is a WR or MIL-STD.

NOTE

Users of MIL-STD-1322 dash number sheets and WR-53 slash number sheets should consult Section 3 of the latest revision of MIL-HDBK-236 to confirm that they are using up-to-date documents.

4.4 Unit load design requirements. All unit loads of ammunition and explosives for domestic and overseas shipment shall meet the design requirements of MIL-STD-1660 for transportability, handling equipment compatibility, shape, size, weight, stability, and stacking capability.

4.4.1 Wood pallets. Unless otherwise specified or it has been determined that wood pallets are undesirable, all palletized unit loads shall be designed utilizing wood pallets.

4.4.2 Steel strapping requirements (Palletized Unit Load). Normally a minimum of four tie-down straps are required per unit load with two straps running the length of the unit load and two straps running the width of the unit load. Unitized loads (loads not requiring a pallet) may have a minimum of two straps. Table I lists the heavy duty strap size authorized for use and the safe working capacity of each strap.

TABLE I. Maximum safe working capacities of steel strapping

Nominal Width and Thickness Size (Inches)	Heavy Duty	
	Cold Rolled Heat Treated (Pounds)	Hot Rolled (Pounds)
3/4 × .031	725	—
3/4 × .035	—	725
1-1/4 × .031	900	—
1-1/4 × .035	—	900

4.4.2.1 Tie-down strap size. The gross weight of a load, divided by the total number of tie-down straps to be used, determines the weight to be borne by each strap. This weight is compared to the strap strength listed in Table I. Strapping which has the same or next higher strength must be used

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4.4.2.2 Horizontal strap size The gross weight of the layer compared with the strap capacities listed in Table I determines the strapping size required.

4.4.3 Steel strapping seal joints All steel strapping seal joints shall be double-notched joints produced and controlled in accordance with the requirements of Appendix B.

4.5 Material Requirements Unless otherwise specified by a dash numbered sheet, material used for the construction of palletized or unitized loads shall be as specified in this document.

4.5.1 Pallets. Four types of pallets are used in the construction of unit loads (a) standard wood pallets (see 4.5.1.1); (b) standard metal pallets (see 4.5.1.2), (c) special purpose wood pallets; and (d) special purpose metal pallets (see 4.5.1.3 and 4.5.1.4)

4.5.1.1 Standard wood pallets. Unless otherwise specified, hardwood species listed in the highest species group number in the following pallet specification shall be used for the construction of standard wood pallets.

Pallet Specification	Deck Size (Inches)	Pallet Type
MIL-P-15011, Style 1 Class 1	40 x 48	Post Construction
NN-P-71, Type V, Size 2	40 x 48	Stringer Construction
MIL-P-15011, Style 1A, Class 1	35 x 45-1/2	Post Construction

4.5.1.2 Standard metal pallets The following standard metal pallets constructed in accordance with MIL-P-23312 shall be used in the construction of unit loads

Pallet	Deck Size (Inches)	Material	Drawing
Mk 3 Mod 0	40 x 48	Steel	10001-564200
Mk 12 Mod 0	35 x 45-1/2	Steel	10001-2086479
Mk 12 Mod 1	35 x 45-1/2	Steel	10001-2645217

NOTE

The Mk 12 Mods 0 and 1 Pallets are functionally interchangeable (see 4.6.3). Strap threading, height and weight will vary (see 4.6.2).

4.5.1.3 Special purpose wood pallets Special purpose wood pallets shall be constructed in accordance with the details on applicable MIL-STD dash number sheet or in accordance with

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drawings specified on the MIL-STD dash number sheet Special purpose wood pallets shall not be used unless specifically authorized by a MIL-STD dash sheet.

4.5.1.4 Special purpose metal pallets. Special purpose metal pallets shall be as specified on the MIL-STD dash number sheet No substitutions are permitted

4.5 2 Palletizing or unitizing adapters

4.5 2 1 Wood palletizing or unitizing adapters Wood palletizing or unitizing adapters, e.g , caps, covers spacer frames, etc shall be constructed in accordance with the details on the MIL-STD dash number sheet

4.5.2.2 Metal palletizing or unitizing adapters Metal palletizing or unitizing adapters shall be as specified on the MIL-STD dash number sheet

4 5.3 Steel strapping All steel strapping shall be new (unused) material in accordance with QQ-S-781, Class 1, Type I or IV, Heavy Duty Finish A or B. The size (width and thickness) of strapping shall be as specified by the MIL-STD dash number sheet

4.5 4 Seals All seals used to join the ends of steel strapping shall be in accordance with QQ-S-781, Type D (double notch), Style I, II, III or IV, Class R or H, Finish A or B The style of seal used shall be selected for compatibility with the tensioning and sealing tools being used The seal class (R or H) shall be compatible to the strapping used

4.5.5 Nails All nails used in the construction of unit loads shall be in accordance with FF-N-105, Type II, and of the style and length specified by the MIL-STD dash number sheet

4 5 6 Staples All staples used to secure strapping to components of the unit load shall be commercial staples sold by strapping manufacturers for use in securing heavy duty strapping of the size (width) strapping being used

4.5 7 Edge protectors Unless otherwise specified, all edge protectors shall be zinc-coated steel of the standard commercial size for the size of strapping used on the unit load.

4.5.8 Wood (lumber). Wood (lumber) used in the construction of unit loads shall be in accordance with MM-L-751 or MIL-STD-731, whichever is required by the MIL-STD dash number sheet When MM-L-751 is cited, the wood shall be number 2 grade or equivalent When MIL-STD-731 is cited, the wood shall be Class 2, structural. Unless otherwise specified, all wood shall be dressed softwood.

4.5.8.1 Selecting lumber. All unitizing lumber should be selected from sound lumber, free from dry rot, knots, knot holes, checks, or splits which will affect its strength or interfere with proper nailing Knots, knot holes, checks, and splits or other defects are permitted in lumber as long as they do not impair the strength of the unit load

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4 5.9 Plywood All plywood used in the construction of unit loads shall be in accordance with NN-P-530, Group B, Type-interior with exterior glue, Grade C-D. The thickness of the plywood shall be as required by the MIL-STD dash number sheet

4 6 General principles of unitizing or palletizing The following general principles should be followed for good unit loads and safe work practices

a. Know the characteristics of the material being unitized and observe all precautions applicable thereto
b. Use the proper unitizing equipment, materials and tools.
c. Examine equipment; e.g., pallets, adapters, etc., to ensure that it is, in all respects, completely suitable for use
d Follow an approved unitizing or palletizing procedure when assembling a unit load
e All components of the unit load (containers or items, adapters frames battens, etc) shall be snug tight and squared up
f Position strapping around the unit load with applicable battens, adapter frames, edge protectors, etc

WARNING

Use only approved handling equipment when handling unit loads, heavy items or containers See OP 4098 and OP 2173.

Stand clear when positioning and tensioning strapping Strapping has sharp corners and edges which are a hazard to unitizing personnel when the ends are inadvertently released during positioning or the strapping breaks during tensioning. Required protective equipment shall be worn by all unitizing personnel when assembling unit loads

g Tension strapping as tight as possible without breaking the strapping or damaging containers or items, metal adapters or wood dunnage components Some crushing of the edges of wood containers and dunnage components is acceptable so long as there is no cracking or splitting of the wood All straps shall be in one piece
h. When necessary, staple strapping to woo dunnage components Do not staple to any containers or lading items.
i Mark all unit loads in accordance with the specified marking requirements of MIL-STD-129.
j Inspect all unit loads in accordance with 4 9 7 to verify the suitability of the unit loads for safe and efficient shipping, handling and storage or stowage

4 6.1 General nailing procedure The proper application of nails will ensure the necessary holding power without the risk of splitting the lumber and affecting the integrity of the unit loads

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Some general rules for nail application which have gained general acceptance are listed below.

a. Except when required, end grain nailing should be avoided. Use sufficient nails. Balanced nailing is important. Stagger nails along the piece being nailed. Do not nail along one grain of wood. Whenever possible, drive nails straight, do not toe-nail unless required by specified construction requirements.
b. Generally, no nail shall be driven closer to the end of piece of lumber than the thickness of that piece, nor closer to the edge than half the thickness of the piece holding the nail head
c. When pieces are of different thicknesses, the nailhead should be in the thinner piece.

WARNING

Never nail directly to the items or containers being unitized.

4.6.2 Strap threading for metal pallets (Mk 3 and Mk 12 Mods 0 and 1). Straps running parallel to the base runner of the pallet shall be threaded between the deck wires of the pallet (see Figure 1) Straps running at right angles to the base runner of the pallet shall pass under the deck wires.

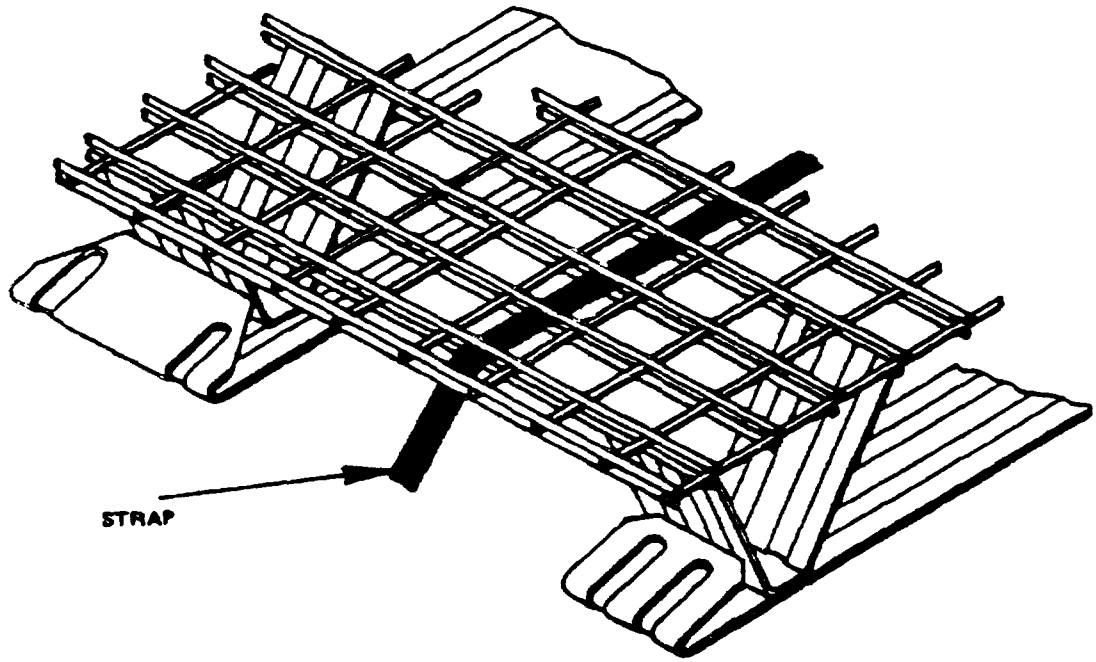


FIGURE 1. Strap threaded between the deck wires of a pallet.

4.6.3 Mk 12 Mod 0 and 1 Pallets The Mk 12 Mod 0 and Mk 12 Mod 1 pallets are interchangeable when constructing unit loads When substituting pallets, care must be taken to thread the straps between the pallet wires so the straps are the same distance from the edge of the unit load

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4.6.3.1 Pallet height. The Mk 12 Mod 1 pallet height is 4-3/4 inches vice 5-1/8 inches for the Mk 12 Mod 0 pallet. When the Mod 1 pallet is used, the overall unit load height will be reduced by 3/8 inch.

4.6.3.2 Pallet weight. The Mk 12 Mod 1 pallet weight will be approximately 72 pounds vice 110 pounds for the Mk 12 Mod 0 pallet. When the Mk 12 Mod 1 pallet is used, the unit load gross weight will be reduced by approximately 38 pounds.

4.7 Unit load marking. In addition to any special marking required by the contract, all unit loads shall be marked in accordance with MIL-STD-129 and the following:

a. The unit load gross weight and cube shall not be marked on items or containers. These markings shall be placed on fiberboard panels or on tags using materials and methods described in MIL-STD-129.
b. All unit loads containing explosives or other hazardous materials shall be marked and labeled in accordance with OP 2165 and 49 CFR 100-199.

WARNING

Use only approved handling equipment when handling unit loads, heavy items or containers. See OP 4098 and OP 2173.

4.8 Deunitizing (depalletizing) procedures. Safe handling and deunitizing of ammunition and explosives is achieved by use of the following general procedures:

a. Move the unit load to a clear area near the point of use or strikedown.
b. If required, square up the unit load so that the unitized items will be stable on the pallet without depending on strapping. Where necessary support items to prevent them from toppling from the pallet. When load is stable, cut strapping.
c. Remove strapping from the unit load and clear the area of cut straps. Strapping should be folded several times and disposed of in a refuse container or other place where it will not be a hazard to personnel or handling equipment and where reuse of the strapping will be prevented.
d. Remove battens, frames, edge protectors, etc. and clear this material from the working area for possible reuse.
e. Remove contents from the unit load.
f. Return to the stock system the metal pallets and pallet adapters for reuse.

4.9 Inspection.

4.9.1 Item to be unitized or palletized. Items to be unitized or palletized shall be free of any visible damage or defects.

4.9.2 Pallets and adapters. Pallets and adapters shall be visually inspected before use for defects that will reduce sturdiness or usability. Pallets and adapters with inadequate or missing welds or

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other signs of damage that will affect their strength or usability shall be rejected If adequate repair is economically feasible, rejected pallets and adapters may be used after the cause for rejection has been corrected

4 9.3 Steel strapping All steel strapping shall be inspected before use for conformance with the size requirements of the MIL-STD dash number sheet and the type, strength (heavy duty) and finish requirements of the general document.

4.9.4 Steel strapping seals. All steel strapping seals shall be inspected before use for conformance with the size requirements of the MIL-STD dash number and the type, style, class and finish requirements of this document

4.9.5 Steel strap tensioning and sealing equipment All steel strap tensioning and sealing equipment shall be inspected before use for compatibility with the required strapping and seal size and any physical defects which will affect the function of the equipment All sealing equipment shall be inspected before and during use for conformance with the capability testing and production control requirements of Appendix B

4 9 6 Accessory material and assemblies. All accessory materials (wood, plywood, fiberboard, edge protectors, nails, etc) and assemblies (trays, caps, top and side frames, spacer frames, etc.) shall be inspected for conformity with (a) 4.5.5 through 4.5.9, (b) the requirements of the MIL-STD dash number sheet or approved palletizing/unitizing plan; and (c) any defect that will affect usage

4.9 7 Unit load The finished unit load or partial unit load (see Appendix A) shall be inspected for conformity to the applicable MIL-STD dash number sheets. Items to be included in this inspection are workmanship, arrangement of items in the unit load, proper selection and positioning of pallet(s) adapter components, and accessory material and assemblies, unobstructed handling equipment access facilities, proper position and tension of strapping, and double notching of seals Markings shall be inspected to assure it is as specified.

4.9.8 Preshipment inspection of unit loads After extended periods of time or numerous handlings the integrity of unit loads can be adversely affected. All unit loads being readied for shipment/transfer shall be inspected and refurbished, if necessary, as required by Appendix C of this document

5 DETAILED REQUIREMENTS

5 1 General Prototypes of unit loads that were documented as MIL-STD-1322 dash numbered sheets have been tested by NWHC to confirm that the design meets all strength and handling requirements Unit loads constructed in accordance with these dash sheets shall be inspected in accordance with the requirements of 4.9

5 1.1 Unit loads constructed in accordance with MIL-STD-1320-000 do not require testing They shall be inspected in accordance with the requirements of 4 9

5 1.2 Unit loads constructed in accordance with NWHC-approved palletizing/unitizing plans do not require testing They shall be inspected in accordance with the requirements of 4.9. The plans are

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approved for one time use only since a specific MIL-STD dash numbered sheet must be developed for recurring requirements

6. Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy — OS
Army — SM
Air Force — 43

Review Activities:
Navy — MC, SH

User Activities:
Navy — AS

Preparing Activity:
Navy — OS

(Project No. 8140-0374)

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APPENDIX A

PARTIAL UNIT LOADS

10 GENERAL

10.1 Scope. This appendix establishes the criteria for the use and the construction of partial unit loads

20 REFERENCED DOCUMENTS

Not applicable.

30 DEFINITIONS

Not applicable

40 GENERAL REQUIREMENTS

40.1 Criteria to use for partial unit loads The use of partial unit loads is to be avoided by filling requisitions/allowances to the nearest full unit load Where insufficient quantities or operational circumstances dictate less than a full unit load, one partial unit load consisting of one or more complete layers of items or containers may be used Quantities of items or containers which do not equal one full layer or are in excess of one or more full layers shall be shipped, handled and stored/stowed loose (not unitized)

40 2 Handling and stowage of loose (not unitized) items The following handling and stowage methods are recommended for loose items or containers

a Dockside loading/offloading. For dockside loading or offloading, use an ammunition pallet crate or a skip box
b Cargo ship stowage Use wood dunnage to retain the items in stowage

50 DETAILED REQUIREMENTS

50 1 Partial unit load construction Except for unit load height (reduced number of layers), the partial unit load shall be constructed in accordance with this document The height of battens, spacer frames, etc shall be reduced to maintain the same relationship with the top surface of the unit load

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APPENDIX B

OPERATION AND PRODUCTION CONTROL OF STEEL STRAP JOINT SEALING EQUIPMENT

10 GENERAL

10.1 Scope. This appendix contains the requirements for the operation and production control of steel strap joint sealing equipment. It also gives the minimum steel strapping joint strength required for ammunition unit loads.

20 REFERENCED DOCUMENTS

Not applicable.

30 DEFINITIONS

Not applicable

40 GENERAL REQUIREMENTS

40.1 General. The integrity of a unit load and, therefore, safety of ammunition handling operations, is dependent upon the strength of the steel strapping holding the unit load together. Since the joint seal of each strap on a unit load is the weakest link in the strap, it is necessary to maintain the required tensile strength of each seal joint produced. To do so, requires close control over the operation (use) and capability of the seal joint notching equipment.

40.2 Operation (use) of equipment

40.2.1 Hand operated equipment (tools). When using hand operated notching tools to produce seal joints, the tool shall be positioned over the seal and actuated until the tool motion has stopped. If the tool is the single notch type, the tool shall be used twice on each seal. Each pair of notches shall be positioned approximately equal distances from each other and the ends of the seal. If the tool double notches (two notches with one stroke of the tool), the notches shall be centered approximately between the ends of the seal.

40.2.2 Power equipment. When using air power tensioning and sealing equipment, the manufacturer's air pressure and lubrication recommendations shall be maintained at all times, including the preparation of test specimens.

40.3 Equipment capability control.

40.3.1 Release to production. Prior to the release of a new or repaired strap sealing tool to production, a sample of five test specimens shall be tested in accordance with 40.6 and evaluated as follows:

a. If all of the test specimens exceed the minimum joint strength specified in 40.4, the tool may be released to production.

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b If all of the test specimens do not meet the minimum joint strength specified in 40 4, the tool shall be adjusted/repaired or replaced and the five specimen test repeated.

40 3.2 Production control procedure. The capability of notching tools to produce acceptable steel strap sealed joints shall be maintained at all times during production use Verification of tool efficiency shall be made in accordance with the following production control schedule and whenever there is any indication that the tool is not producing acceptable joints

40.3.2.1 Verification procedure. The capability of a notching tool shall be verified in accordance with the following procedure.

a Prepare samples of three test specimens at random from a unit load and test them in accordance with 40.6
b If all of the test specimens exceed the minimum joint strength specified in 40.4, the tool may be used for further production of unit loads until the next time verification is required.
c If all of the test specimens do not meet the minimum joint strength specified in 40 4, the tool shall be returned to maintenance for repair

40 3.2.2 Producuon control schedule The capability of notching tools shall be verified in accordance with the following schedule

a. Whenever a seal notching tool is put to use in production, a sample of notched seal joints shall be taken from the first unit load produced and subjected to the verification procedure of 40.3.2.1.

b During production, if there is no indication that the seal notching tool is not producing satisfactory joints, a sample of notched seal joints shall be taken from a unit load and subjected to the verification procedure of 40.3.2.1 at intervals of approximately 250 joints

c When power seal notching tools are used in areas where the production rate does not provide for one sample per shift, a sample shall be taken once per shift.

d. When manually operated seal notching tools are used in low production areas where the production rate does not provide for one sample per month, a sample shall be taken at least once a month

40 4 Minimum notched seal joint strength. Minimum notched steel strap seal joint strengths for steel strapping used on ammunition and explosive unit loads are as follows:

	Minimum Joint Strength	
Strap Size (inches)	Pounds	Kilograms
3/4 × 035 or .031	1900	860
1-1/4 × .035 or .031	3000	1360

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40.5 Test specimen. All test specimens shall consist of the same notched seal-strap combination to be used in the production of unit loads. Each test specimen shall have unaltered strapping (not bent, notched, etc.) adjoining the notched seal joint and shall be free of any welded joint in the strapping. The length of the test specimen shall be at least 18 inches overall and at least 12 inches between the tensile tester grip areas. The joint shall be midway between the tester grip areas.

40.6 Tensile test. Test specimens in accordance with 40.5 shall be subjected to a tensile test in a tensile testing machine capable of performing at the loads and speed rate required. Normally, such a device is available in a commercial or Government quality evaluation laboratory. However, where large quantities of unit loads are being assembled some distance away from the laboratory, use of a locally fabricated small tensile tester is authorized. Drawings for such a device are available from:

Commanding Officer
Naval Weapons Station Earle (Code 80A)
Colts Neck, NJ 07722

40.6.1 Test procedure. Place the test specimen in the grips of a calibrated tensile tester so that the load is transmitted axially to the joint of the specimen. Activate the tensile tester and apply the load to the test specimen until the joint fails. A tensile tester pull rate of approximately 0.2 inch/minute should be used.

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APPENDIX C

PRESHIPMENT INSPECTION AND REFURBISHMENT REQUIREMENTS FOR UNIT LOADS IN SERVICE

10. GENERAL

10.1 Scope This appendix gives guidelines for preshipment inspection and refurbishment, if necessary, of unit loads of ammunition and explosives for domestic and overseas shipment to assure the safety and adequacy for handling and shipment after possible degradation over extended periods of storage.

20. REFERENCED DOCUMENTS

Not applicable

30. DEFINITIONS

Not applicable.

40. GENERAL REQUIREMENTS

40.1 General The safety of ammunition and explosive handling operations is dependent upon the integrity of the unit load After extended periods of time or numerous handling operations, the integrity of unit loads can be adversely affected The condition of each unit load must be inspected, and refurbishment performed as necessary to assure the safety of subsequent shipping and handling operations Details of each situation such as transportation mode, subsequent handling of the shipment, and the end use (disposition) of the material should be weighed in establishing the functional adequacy in accordance with the guidelines provided in the detailed requirements section of this appendix.

40.2 Performance Inspection and refurbishment, if required, in accordance with this appendix, shall be performed as part of the preparation for shipment or transfer of hazardous materials regardless of the condition of the unit load. All accessible areas shall be examined to determine the weakest part of the load

50 DETAILED REQUIREMENTS

50.1 Strap corrosion Strapping on unit loads shall show no reduction in strap width, ragged strap edges or heavy rust scale with readily visible pitting (scale should be removed in a small area by scraping to inspect for pitting). Strapping of questionable strength must be replaced

50.2 Strap looseness

50.2.1 Inspection method. Unit load strapping shall be inspected for looseness as follows. Pull side strapping (vertical area) away from load by hand, without using body weight Measure distance

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strap moves away from unit load (do not measure from indented area of load) If it can be moved more than 1-1/2 inches, strap must be retightened, or replaced as follows.

a. Steel strapping which has been applied and tensioned around metal items, containers, metal edge protectors, bomb pallets, metal pallet adapters, or metal hardware on wood containers shall not be retensioned but shall be replaced when loose.
b Steel strapping which has been applied to any unit load and tensioned, shall not be retensioned by splicing in additional strapping or removed and reused on any other unit loads.
c. Steel strapping which has been tensioned and sealed on a unit load of wood containers or a unit load where the strapping is around wood adapters or wood edge protectors, may be retensioned by the method shown in Figure 2. Except for the standard metal pallets, strapping shall not have corner bends around any metal objects when this method is used When retensioning, the corner bends in the strapping shall be in the same position they were in when the strapping was originally tensioned. Strapping which has been damaged in any way shall not be retensioned but shall be replaced

50 3 Pallets and adapters

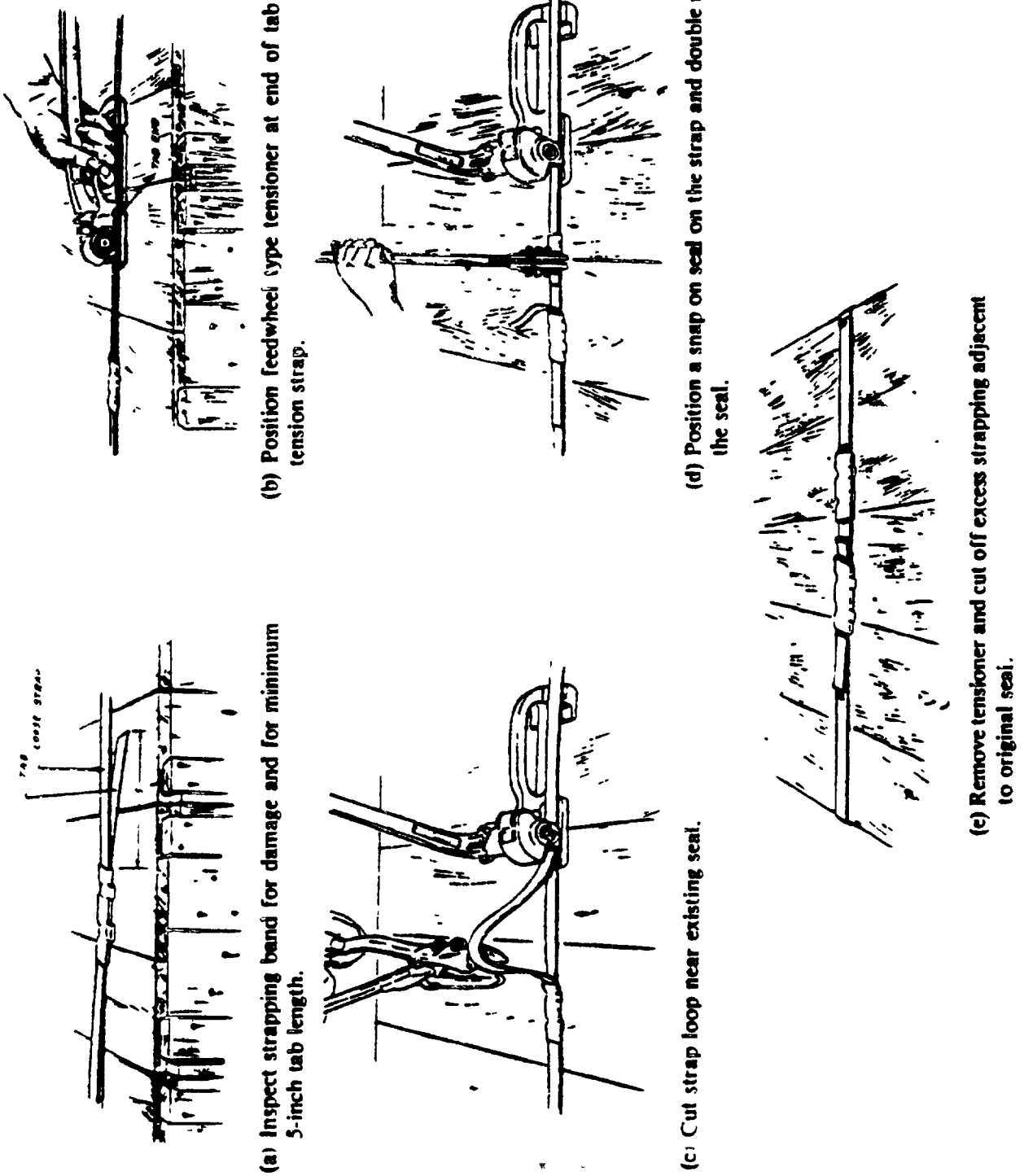
50 3 1 Standard metal pallets Standard metal pallets shall be inspected to the extent possible in accordance with OR-99/88P1000

50.3.2 Special purpose metal pallets. Special purpose metal pallets shall be inspected to the extent possible for defects that will reduce sturdiness or useability Pallets with inadequate or missing welds or other signs of damage that will affect their strength or useability shall be rejected If adequate repair is economically feasible, rejected pallets may be used after repair. OR-99/88P1000 may be used for guidance.

50.3.3 Metal pallet adapters. Metal pallet adapters shall be visually inspected for any defects which will reduce sturdiness or useability Adapters with inadequate or missing welds or other signs of damage that will affect their strength or useability, shall be rejected If adequate repair is economically feasible, rejected adapters may be used after repair.

50.3.4 Wood pallet adapters. Wood pallet adapters shall be visually inspected for defects that will reduce sturdiness or useability Adapters with loose, broken, or missing boards, or protruding or missing nails, shall be rejected If adequate repair is economically feasible, rejected adapters may be used after repair

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(a) Inspect strapping band for damage and for minimum 5-inch tab length.

(b) Position feedwheel type tensioner at end of tab and tension strap.

(c) Cut strap loop near existing seal.

(d) Position a snap on seal on the strap and double notch the seal.

(e) Remove tensioner and cut off excess strapping adjacent to original seal.

FIGURE 2 Strap retensioning

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DOCUMENT IDENTIFIER AND TITLE
MIL-STD-1322A UNIT LOADS OF AMMO & EXPLOSIVES FOR DOMESTIC & OVERSEAS SHIPMENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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